SQL Assignment: Sales Database

Keys • DDL • DML • Constraints

Name: _____ Date: _____

Batch: _____ Marks: / 25

Scenario: A retail company wants a database to track its **Customers**, **Products**, **Sales Orders** and **Order Items**. Use MySQL 8 syntax for all answers.

Instructions: Write all queries in a single `.sql` file, attach an output screenshot for each query, and write theory answers as comments or in a separate document.

PART 1: Table Design & Constraints

Q1. Create the table `Customer`: [1 mark]

- `CustomerID`: PRIMARY KEY, AUTO_INCREMENT; `FullName`: NOT NULL
- `Email`: UNIQUE and NOT NULL; `Phone`: UNIQUE
- `Age`: CHECK 18 or above; `City`: DEFAULT 'Kochi'

Q2. Create the table `Product`: [1 mark]

- `ProductID`: PRIMARY KEY, AUTO_INCREMENT; `ProductName`: NOT NULL; `SerialNumber`: UNIQUE
- `Brand`: DEFAULT 'Dell'; `Price`: CHECK (Price > 0); `StockQty`: CHECK (StockQty >= 0)

Q3. Create `SalesOrder` and `OrderItem`: [2 marks]

- `SalesOrder`: `OrderID` PRIMARY KEY AUTO_INCREMENT; `CustomerID` foreign key with ON DELETE CASCADE; `OrderDate` NOT NULL, defaults to the current date; `Status` CHECK in ('Pending','Shipped','Delivered','Cancelled'), DEFAULT 'Pending'
- `OrderItem`: `(OrderID, ProductID)` as a **composite primary key**; `OrderID` foreign key with ON DELETE CASCADE; `ProductID` foreign key with ON DELETE RESTRICT; `Quantity` CHECK (Quantity > 0); `UnitPrice` CHECK (UnitPrice > 0)

Q4. Using `ALTER TABLE`, add a column `DiscountPercent` (DEFAULT 0, CHECK between 0 and 50) to `OrderItem`, add a CHECK on `Customer.Phone` so it has exactly 10 digits, and drop the UNIQUE constraint on `SerialNumber`. [1 mark]

PART 2: INSERT & Constraint Violations

Q5. Insert at least **5 customers** and **5 products** (for example: Laptop/HP/55000, Mouse/Logitech/1200, Keyboard/Dell/2500). Include one customer with no `City` and one product with no `Brand`, so the DEFAULT values apply. Show the result using SELECT. [1 mark]

Q6. Insert at least **6 sales orders** and **10 order items**, where some customers place multiple orders and some orders contain multiple products. [1 mark]

Q7. Try each insert below. For each, write the **error message** and the **constraint** that caused it: [2 marks]

- A customer with a duplicate email
- A product with Price = 0, and another with a negative StockQty
- A duplicate (OrderID, ProductID) order item
- An order for a CustomerID that does not exist
- An order with Status = 'Returned'

PART 3: UPDATE & DELETE

Q8. Write UPDATE queries to: [1.5 marks]

- Increase the price of all products of one brand by 10%
- Change the status of one specific order to 'Shipped'
- Set the city to 'Trivandrum' for all customers whose city is NULL (hint: insert one customer with an explicit NULL city first)

Q9. Write DELETE queries to delete one order item, and then delete all orders with Status 'Cancelled'. Check what happened to the items of those orders. [1 mark]

Q10. Delete a customer who has orders. Show `SalesOrder` and `OrderItem` before and after, and explain what happened to the related rows (CASCADE). [1.5 marks]

Q11. Try deleting a product that has already been sold. Write the error you get and explain how ON DELETE RESTRICT caused it. Then write the steps to remove that product successfully. [1 mark]

PART 4: DELETE vs TRUNCATE vs DROP (Practical)

Q12. Run `TRUNCATE TABLE Customer;` and `DROP TABLE Customer;` while `SalesOrder` still references it. Write the error for each and explain why. [1 mark]

Q13. Create a copy table `Product_Backup` (using `CREATE TABLE ... LIKE` and `INSERT ... SELECT`). Then check how AUTO_INCREMENT behaves: [2 marks]

- Run DELETE (without WHERE) on the copy and insert a new row. Does the counter continue or reset?
- Refill the copy, run TRUNCATE, and insert a new row. Does the counter continue or reset?
- Explain the difference you observed.

Q14. Write the correct order of commands to completely remove all tables (including the backup) without errors, and explain why the order matters. [1 mark]

Q15. Scenario: A sales manager accidentally ran `DELETE FROM SalesOrder;` and another employee ran `TRUNCATE TABLE SalesOrder;`, both inside open transactions. Which one can be recovered using ROLLBACK, and why? Which is faster? Which would you use to remove only cancelled orders? What extra danger does the first command carry in this database? [1.5 marks]

PART 5: Keys & Theory

Q16. Given `Customer(CustomerID, Email, Phone, FullName, Age, City)`, where CustomerID, Email and Phone are each unique: [1.5 marks]

- List at least **5 super keys**
- Identify all **candidate keys** and choose the **primary key** with a reason
- List all **non-prime attributes**

Q17. In `OrderItem`, explain with an example why neither `OrderID` alone nor `ProductID` alone can be the primary key. Why is a composite key the right choice here? [1.5 marks]

Q18. Answer Yes or No with a short example: Can a table have (a) more than one candidate key, (b) more than one primary key, (c) more than one foreign key (hint: look at OrderItem), (d) a foreign key referencing a UNIQUE (non-primary) column? [1.5 marks]

Q19. Can a foreign key contain NULL or duplicate values? Can a primary key? Explain the difference between PRIMARY KEY and UNIQUE in your answer. [1 mark]

Total: 25 marks | Neat formatting, meaningful names and a screenshot for every query carry marks.